SA-LuT-Nets: Learning Sample-Adaptive Intensity Lookup Tables for Brain Tumor Segmentation

Biting Yu, Luping Zhou, *Senior Member, IEEE*, Lei Wang, Wanqi Yang, Ming Yang, Pierrick Bourgeat, and Jurgen Fripp

***Abstract*— In clinics, the information about the appearance and location of brain tumors is essential to assist doctors in diagnosis and treatment. Automatic brain tumor segmentation on the images acquired by magnetic resonance imaging (MRI) is a common way to attain this information. However, MR images are not quantitative and can exhibit significant variation in signal depending on a range of factors, which increases the difficulty of training an automatic segmentation network and applying it to new MR images. To deal with this issue, this paper proposes to learn a sample-adaptive intensity lookup table (LuT) that dynamically transforms the intensity contrast of each input MR image to adapt to the following segmentation task. Specifically, the proposed deep SA-LuT-Net framework consists of a LuT module and a segmentation module, trained in an end-to-end manner: the LuT module learns a sample-specific nonlinear intensity mapping function through communication with the segmentation module, aiming at improving the final segmentation performance. In order to make the LuT learning sample-adaptive, we parameterize the intensity mapping function by exploring two families of non-linear functions (i.e., piece-wise linear and power functions) and predict the function parameters for each given sample. These sample-specific parameters make the intensity mapping adaptive to samples. We develop our SA-LuT-Nets separately based on two backbone networks for segmentation, i.e., DMFNet and the modified 3D Unet, and validate them on BRATS2018 and BRATS2019 datasets for brain tumor segmentation. Our experimental results clearly demonstrate the superior performance of the proposed SA-LuT-Nets using either single or multiple MR modalities. It not only significantly improves the two baselines (DMFNet and the modified 3D Unet), but also wins a set of state-of-the-art segmentation methods. Moreover, we show that, the LuTs learnt using one segmentation model could also be applied to improving the performance of another segmentation model, indicating the general segmentation information captured by LuTs.**

Manuscript received November 13, 2020; revised January 18, 2021; accepted January 27, 2021. Date of publication February 3, 2021; date of current version April 30, 2021. This work was supported in part by the Australian Research Council under Grant ARC DP200103223 and in part by the National Science Foundation of China under Grant 61876087 and Grant 62076135. *(Corresponding authors: Luping Zhou; Lei Wang.)*

Biting Yu and Lei Wang are with the School of Computing and Information Technology, University of Wollongong, Wollongong, NSW 2522, Australia (e-mail: leiw@uow.edu.au).

Luping Zhou is with the School of Electrical and Information Engineering, The University of Sydney, Sydney, NSW 2006, Australia (e-mail: luping.zhou@sydney.edu.au).

Wanqi Yang and Ming Yang are with the School of Computer Science and Technology, Nanjing Normal University, Nanjing 210023, China.

Pierrick Bourgeat and Jurgen Fripp are with CSIRO Health and Biosecurity, Brisbane, QLD 4001, Australia.

Digital Object Identifier 10.1109/TMI.2021.3056678

***Index Terms*— Segmentation, magnetic resonance imaging (MRI), brain, neural network.**

I. Introduction

THE location and appearance of brain tumors are key to the diagnosis and treatment of brain cancers. Such information is usually acquired by the non-invasive magnetic resonance imaging (MRI) and extracted by segmenting the tumor regions in the scanned images. However, accurate brain tumor segmentation is always challenging. For example, as one of the most aggressive brain tumors, glioma affects tens of thousands adults around the world [1]–[3]. Compared with some brain tumors like meningiomas, the extent of gliomas is more difficult to accurately define, due to their different shapes, sizes, and diffused locations which vary from patient to patient [4]. Moreover, gliomas often extend their tentacle-like structures to invade the healthy brain tissues rather than just replacing them, which results in subtle changes and fuzzy tumor boundaries [5]. The accurate manual annotations of gliomas require laborious efforts from the professional radiologists [6] and could burden the workload in clinics. Therefore, automatic brain tumor segmentation methods on MR images would be beneficial in speeding up the process and provide objective and repeatable measurements for the radiation therapy treatment.

To improve the segmentation accuracy of brain tumors, various machine learning based methods have been investigated to analyze MR images. Among these methods, convolutional neural networks (CNNs) based models are especially prevalent due to their efficiency and effectiveness [4], [7]–[14]. Recently, a 2D CNN model, Unet [15], has been widely explored and made into diverse 3D architectures to globally grasp the content of 3D objects from MR images and shown promising performance for brain tumor segmentation. For example, the model, NVDLMED, additionally employed an image reconstruction branch with the original segmentation branch in the 3D Unet to impose more constraints on the encoding layers and raised the segmentation accuracy [16]; The modified 3D Unet cooperated residual layers with the skip connections in its structure to enable more communications among both neighboring and distant MR image features and better locate tumors [17], [18]; DMFNet integrated multi-fiber units with dilated layers into the original 3D Unet architecture to extract features within

1558-254X © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

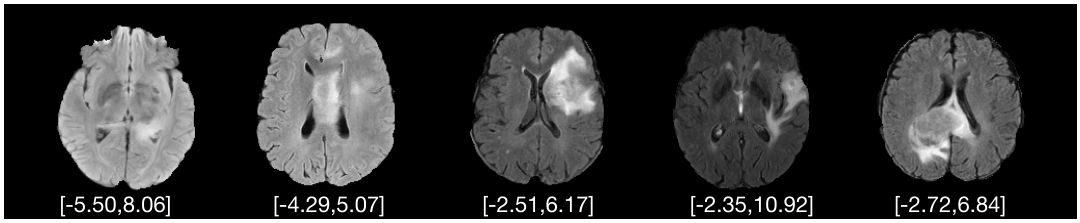

Fig. 1. Tumor carried MR images of FLAIR modality preprocessed after the zero-mean and unit-STD normalization. Their normalized intensity scales are given in the bottom of images. After this linear rescaling, the significant intensity variation among MR images still remains.

multi-scale scope [19]; S3S-Unet applied novel three-branch units to convolutional layers with fewer trainable parameters for the higher efficiency of image inference [20]. All the above-mentioned models use more effective CNN-based architectures and achieve impressive brain tumor segmentation performance.

The intensity values in MR images are not quantitative. Even the scanned intensities of the same tissue type can be distinctly different among MR images [21]–[24]. In the brain tumor imaging, this results in the tumor-surrounding contrast being much weaker in some MR images than in the others. This intensity variation inevitably increases the difficulty of training the segmentation model and generalizing it to new MR images, which is the problem of interest in this work. To eliminate variations in image intensity, a preprocessing step, intensity normalization [22], is used to align the intensity values of MR images with a standard before image analysis is performed. Although it (intensity normalization) has been seen sporadically in some automatic MR image segmentation methods (e.g., [13]), it is not widely applied in the recent deep learning based segmentation. This is because intensity normalization and segmentation are two kinds of tasks with very different nature. The former usually alters the histograms of MR images so that the discrepancy among these histograms is minimized after the normalization step [22], [23], [25]. This operation may undesirably erase critical pathological clues in the original MR images and then adversely affect the segmentation performance. Instead of complicated intensity normalization steps, a simple zero-mean and unit-STD (standard deviation) normalization is often adopted to preprocess MR images, which linearly rescales the intensity values. However, as shown in Fig 1, the intensity values still exhibit significant variation after this normalization.

This paper proposes an end-to-end learning framework where sample-adaptive intensity lookup tables (LuTs) are learnt to cope with differently contrasted MR images and promote brain tumor segmentation performance. The idea behind our work comes from our observation about how doctors deliver manual annotation. Relying on their expertise, the doctors usually adjust the intensity contrast of the whole MR image using software tools to make the regions of interest protruded, so that the contours of these regions could be better observed. During this process, the doctors attempt to find a relatively ideal intensity mapping function through their naked (but trained) eyes for annotation. This procedure highly depends on the doctors' individual experience and the images are processed one by one. Inspired by this observation, we propose a learning framework that has two benefits. On one hand, it automatically learns the intensity-level mapping function (i.e., LuT) that suits the following segmentation task, which does not need the doctors' intervention; on the other hand, the learnt mapping function adapts to various MR images with different tumor-surrounding contrasts, therefore relaxing the training and generalization of the segmentation model to some extent. That is, each individual MR image is particularly assigned with an intensity LuT, which is associated with a nonlinear mapping function that caters for the individual need of intensity adjustment. We explore two families of nonlinear mapping functions, i.e., piece-wise linear mapping functions and power mapping functions. The parameters of the mapping function are automatically predicted in our SA-LuT-Net, which vary with the input MR images. The intensity LuT and the subsequent segmentation model are trained together in an end-to-end manner only with the supervision of segmentation criterion. In this way, they can negotiate with each other to improve the segmentation performance. Some previous works have applied adaptive window based approaches on medical images. However, they are fundamentally different from our work. For example, [26] proposes a task-adaptive approach to modify intensity values for classification, which learned a single window function that is uniformly applied to all training and test images, as most existing deep learning models did. This is different from our sample-adaptive strategy that differently adjusts the contrast of each image for segmentation. The work in [27] estimates the adaptive sizes of local windows for contour points during level-set based segmentation, differing from our method fundamentally in motivation, designed solution, and applicable scenarios.

The contributions of our work are highlighted as follows.

1) An end-to-end learning framework SA-LuT-Net is proposed to jointly learn the sample-adaptive LuTs and the brain tumor segmentation task from MR images. The learnt LuTs vary with the need of the input MR images for the intensity adjustment towards better segmentation. In this way, when new unseen samples whose intensity distributions are different from the training set arrive, our SA-LuT-Net could predict their specific optimal LuTs to transform their intensities for the segmentation. Also, it is also noteworthy that our LuT-transformed images are learnt without requiring any auxiliary information beyond the original segmentation labels.
2) Two families of nonlinear mapping functions, i.e., the piece-wise linear functions and the power functions, are explored to parameterise the intensity mapping to be learnt in the proposed framework. Their effectiveness is experimentally validated and discussed.

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

3) The proposed framework is developed upon two backbone networks, respectively, i.e., the modified 3D Unet [17] and DMFNet [19], which achieved the state-of-the-art performance on brain tumor segmentation. The effectiveness of our framework is verified in both single- and multi-modality segmentation cases.
4) The performance of our SA-LuT-Net is validated on both BRATS2018 and BRATS2019 [6] validation sets. The online evaluation results demonstrate the superior performance of our method over many other state-of-the-art methods, while using fewer model parameters.
5) We also show that, the LuTs learnt using one segmentation model could be transferred to another segmentation model to improve the latter's performance. This suggests that some general information about how to adjust the intensity levels for the segmentation task, rather than for a specific segmentation model, has been learnt.

Our preliminary work was published in the conference paper [28]. Compared with it, this paper has been significantly extended in the following aspects.

1) We explore more nonlinear mapping functions for the intensity mapping. In [28], only the piece-wise linear functions were studied. In this paper, we also investigate using power functions to learn the intensity mapping of the LuTs and validate them on two backbone models, i.e., the modified 3D Unet and the DMFNet.
2) In this paper, we investigate whether the learnt LuTs could be general, to some extent, to the segmentation tasks, rather than being segmentation-model-specific. Therefore, we study if the LuTs learned using one segmentation model (say, modified 3D Unet/DMFNet) could improve the performance of another segmentation model (say, DMFNet/modified 3D Unet) and show promising performance.
3) We verify the effectiveness of our proposed SA-LuT-Nets on more datasets in this paper. Specifically, we test on the BRATS2019 validation set and compare the results with more existing models.
4) We provide more qualitative analysis in this paper. Visual comparisons between different intensity mapping functions (piece-wise linear functions vs power functions), SA-LuT-Nets and the intensity normalization method, and the examples from the ablation study are provided, which is missing in [28].

II. Proposed Method

This paper proposes a SA-LuT-Net framework to explicitly handle MR intensity variation for segmentation through jointly learning sample-adaptive intensity LuTs and segmentation. An intensity LuT corresponds to a nonlinear mapping function that could be used to adjust the intensity levels of MR images from one set to another. In our case, an LuT takes the intensity levels from the given MR image as the input, and outputs the transformed intensity levels that adjust the input contrast in the given MR image. In this way, the relevant visual information, like pathological tissues, could be better viewed. Due to the varying requirement of intensity adjustment from image to image, an input MR image may need a sample-specific mapping function and hence the corresponding LuT should be adaptive to different input image. In the proposed SA-LuT-Net, the learning of the sample-adaptive LuT is guided by the segmentation performance so that the learnt LuT could match the segmentation task. Fig. 2 illustrates the overview of our proposed SA-LuT-Net framework. To be more specific, under the single-modality scenario, our framework incorporates one LuT module and one subsequent segmentation module. The LuT module generates the sample-adaptive parameters that determine different mapping functions of the LuTs. Each mapping function is then applied to the input MR image to change its intensity contrast. After that, the LuT-transformed image is further used as the input of the following segmentation module to predict the final segmentation labels. Through the end-to-end training in the proposed SA-LuT-Net, the LuT and the segmentation modules are jointly trained to negotiate with each other and achieve the optimal segmentation results.

A. Intensity LuT Module

In order to make the LuT sample-adaptive, we parameterise the intensity mapping function to be learnt. Diverse nonlinear functions could be used to model the intensity transformation mappings of LuTs. Considering both the simplicity and flexibility, we choose two families of non-linear functions, i.e., power functions and piece-wise linear functions, in this work.

1) Power Function: Power functions are investigated as the nonlinear intensity mapping function associated with LuTs, which are defined as follows and illustrated in Fig. 3 (a).

$$\hat{\mathbf{x}} = \mathbf{x}^{\gamma}, \quad \text{where} \quad 0 \leq \mathbf{x} \leq 1, \tag{1}$$

where the intensity levels in the given MR image and its LuT-transformed image are indicated by $\mathbf{x}$ and $\hat{\mathbf{x}}$, respectively, and the power γ is the only learnable parameter that controls the LuT curve for each input MR image. It is noted that, the real input intensity levels are linearly rescaled in each MR image, so that both $\mathbf{x}$ and $\hat{\mathbf{x}}$ are between $[0, 1]$. The curve of the power function passes through the two points $(0, 0)$ and $(1, 1)$, and here we require $\gamma > 0$. When $\gamma = 1$, there will be no intensity adjustment, shown as the dashed diagonal line in Fig. 3 (a). When $\gamma > 1$, the power function has a convex shape for $\mathbf{x} \in [0, 1]$; when $0 < \gamma < 1$, the power function has a concave shape for $\mathbf{x} \in [0, 1]$. Fig. 3 (a) also gives two LuT curves of $\gamma = 2$ and $\gamma = 1/2$ as examples. Power functions are also used in gamma correction, as a nonlinear operator to correct the brightness values of the pixels in natural image photographing and the display of medical images [29]–[35].

In the proposed SA-LuT-Net, the parameters of mapping functions are estimated by a LuT module. When applying the power functions to the intensity adjustment, the LuT module processes a given input MR image and predicts the sample-adaptive value of the parameter γ for this input image. Our LuT module consists of three convolutional blocks and three fully connected (FC) blocks, as shown in Fig. 4. Each convolutional block sequentially consists of a convolutional layer that applies the kernels of size $4 \times 4 \times 4$ and the stride of four, an instance normalization layer, and also a LeakyReLU

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

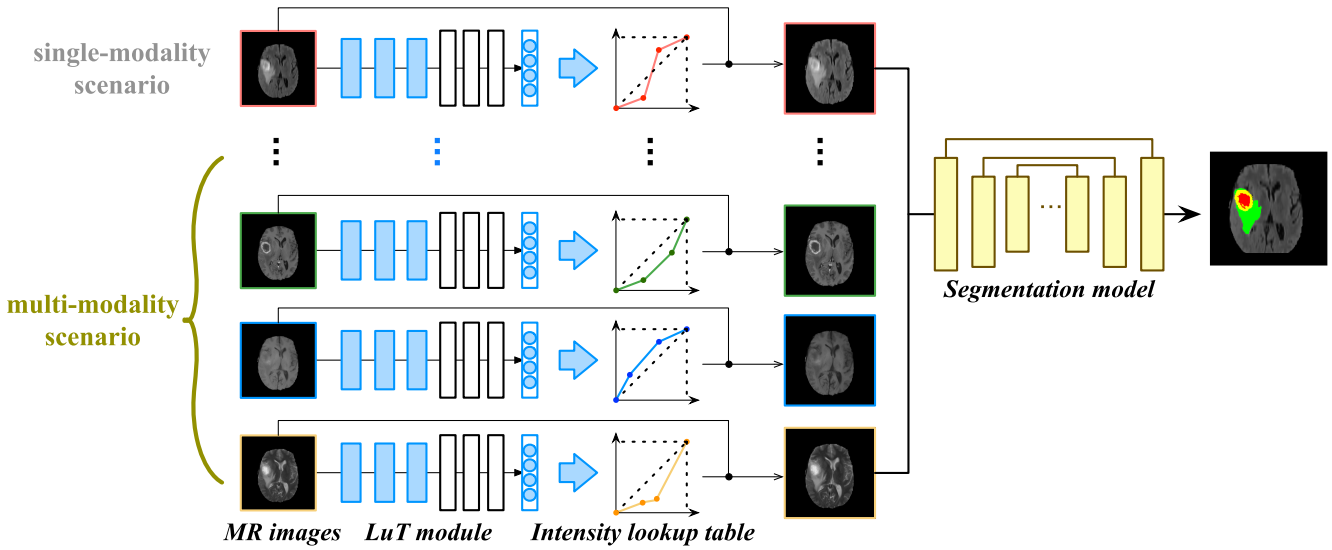

Fig. 2. Overview of the proposed SA-LuT-Net framework under single-modality or multi-modality scenario. For the single-modality scenario, it integrates two modules into the joint learning: (1) a LuT module generating the particular parameters of intensity mapping functions for every input MR image, and (2) a segmentation module processing the intensity adjusted MR images to estimate the labels of tumor regions. In this way, the learnt LuTs could help the input MR images become more suitable for the downstream segmentation task and improve the ultimate segmentation accuracy.

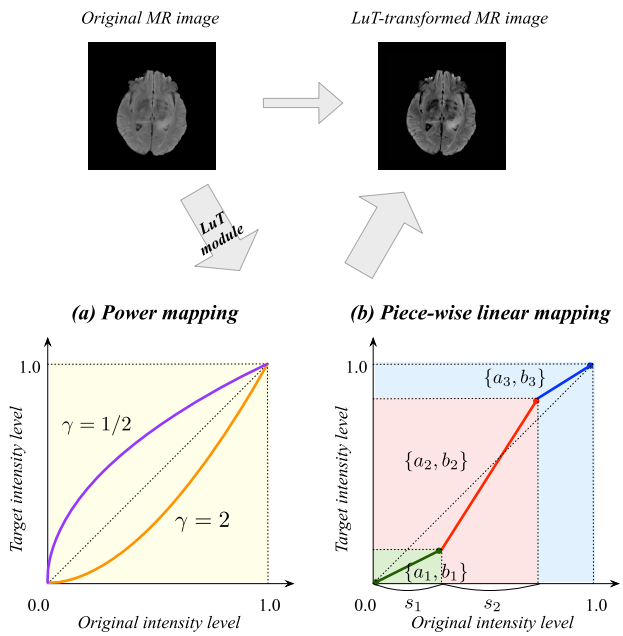

Fig. 3. Two intensity lookup tables separately using (a) a power function and (b) a piece-wise linear function. Both of them can transform the input intensity levels in an MR image to the target levels by the estimated sample-specific parameters of mapping functions.

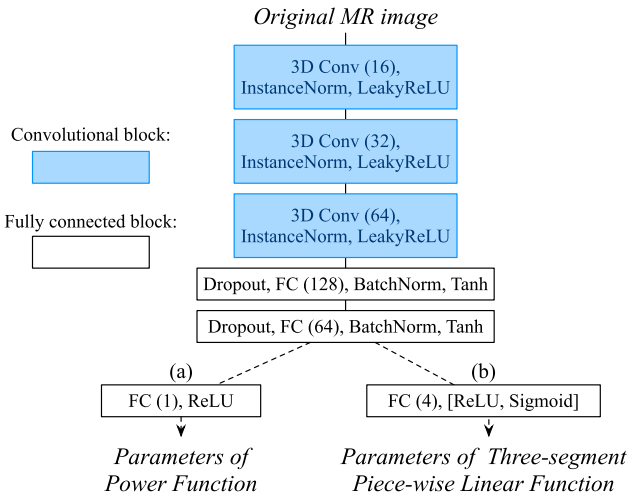

Fig. 4. The architecture of a LuT module. It includes three convolutional blocks and three FC blocks to predict the parameters of the LuT. In its last block, it uses different activation functions to constrain the value ranges of the parameters for (a) power function or (b) three-segment piece-wise linear function.

layer with the slope 0.2. These three convolutional blocks have 16, 32, and 64 output channels in order. In the first two FC blocks, a dropout layer using the rate of 0.5 is first employed, followed by an FC layer, a batch normalization layer, and a Tanh layer. The outputs of these two FC blocks have 128 and 64 channels, respectively. For the last FC block, an FC layer that outputs one channel is applied. Besides, in order to build a regular one-to-one mapping function, we require $\gamma > 0$ according to the definition. Therefore, the ReLU function is then applied in the last FC block to enforce it to be a non-negative value for the LuT transformation.

2) Piece-Wise Linear Function: A three-segment piece-wise linear function is plotted in Fig. 3 (b) as an example, and it could be mathematically formulated as:

$$\hat{\mathbf{x}} = \begin{cases} a_1\mathbf{x} + b_1, & 0 \leq \mathbf{x} < s_1, \\ a_2\mathbf{x} + b_2, & s_1 \leq \mathbf{x} \leq (s_1 + s_2), \\ a_3\mathbf{x} + b_3, & (s_1 + s_2) < \mathbf{x} \leq 1. \end{cases} \tag{2}$$

The slope and the bias of the i-th line segment are denoted by a_i and b_i $(i = 1, 2, 3)$, respectively, while the horizontal intervals of the i-th line segment are s_i $(i = 1, 2)$. In a special case, when $a_i = 1$ and $b_i = 0, \forall i$, the LuT becomes an identity mapping and the intensity levels of the given MR image will

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

remain the same. The identity mapping function is illustrated as the dashed straight diagonal line in Fig. 3 (b), indicating no intensity adjustment.

A LuT module is also employed to predict the sample-specific values of the parameters a_i, b_i, and s_i for each input image. Taking the three-segment piece-wise linear mapping as an instance, our LuT module will output a four-dimensional vector, including a_1, a_3, s_1, and s_2 as its elements. In addition, according to the definition of mappings, the first line segment starts from the origin so that the parameters $b_1 = 0$, and other parameters a_2, b_2, and b_3 can be calculated from the output parameters of the LuT module. Combining the above together, Equation (2) can be rewritten as:

$$\hat{\mathbf{x}} = \begin{cases} a_1\mathbf{x}, & 0 \leq \mathbf{x} < s_1, \\ \dfrac{a_3(s_1+s_2-1) - a_1 s_1 + 1}{s_2}\mathbf{x} \\ \quad + \dfrac{(a_1 - a_3)(s_1+s_2)s_1 + (a_3-1)s_1}{s_2}, & s_1 \leq \mathbf{x} \leq (s_1+s_2), \\ a_3\mathbf{x} - a_3 + 1, & (s_1+s_2) < \mathbf{x} \leq 1. \end{cases} \tag{3}$$

Compared with the power functions, these three-segment piece-wise linear functions have more parameters to learn, so may also have more flexibility for the adjustment.

For the piece-wise linear functions, we also use the architecture of the LuT module in Fig. 4 to predict the parameters a_1, a_3, s_1, and s_2, except that the number of the output channel in the last layer of LuT module becomes four and different nonlinear activation functions are used to predict the four target parameters. To be specific, as $a_1 > 0$ and $a_3 > 0$ are defined to be larger than zero, the last FC block uses the ReLU function to guarantee the positive values of them. Also, s_1 and s_2 sit between zero and one according to definition. Thus, a Sigmoid function is applied to ensure s_1 and s_2 in the range of [0, 1].

3) Multi-Modality Scenario: For the single-modality segmentation task, one LuT module using either the three-segment piece-wise function or power function possesses about 0.57M trainable parameters. If using multi-modality MR images, each modality will be associated with a LuT module. This design considers that separately using different LuT modules for each input modality helps to focus on the specific contrast characteristics of each modality for the segmentation. Therefore, our learnt LuT adapts to different samples and different modalities. Besides, these multiple LuT modules are jointly trained with the subsequent segmentation module. In this way, different LuT modules can communicate well with each other and collaborate together to adjust their corresponding modality images for the final segmentation task.

B. Segmentation Module

In the right part of Fig. 2, our segmentation module is illustrated. Compared with those stand-alone segmentation models, ours takes the intensity-adjusted MR images as its input and is jointly trained with the LuT module. Here, our SA-LuT-Net framework is developed upon two backbone networks, respectively, which are two state-of-the-art models for brain tumor segmentation. The first one is the modified 3D Unet model [17], which contains 26 convolutional layers. It incorporates residual connections in its encoding convolutional blocks to combine the neighboring shallow and deep features, and also integrates different-depth segmentation outputs in an element-wise summation way to apply deep supervision for the final model output. The modified 3D Unet achieved the third-place in BRATS2017 [6]. The second backbone network is DMFNet [19]. It has 69 convolutional layers built as a 3D Unet-like structure with skip connections. It replaces the ordinary convolutional layers in its first six encoding residual units with more efficient adaptive dilated multi-fiber layers to capture the multi-scale feature representations from brain tumor images. It attained the comparable results on the BRATS2018 validation set [6] with the challenge first-place model, NVDLMED [16], but only includes about 1/10 learning parameters of NVDLMED. As for the detailed architectures of the modified 3D Unet and DMFNet, please be referred to the original papers [17] and [19].

C. Training Strategy

Similar to the common segmentation models, the proposed SA-LuT-Net only exploits the ultimate segmentation loss to guide its learning process. Since the LuT module is in the beginning layers (close to the input) of the entire model and the following segmentation module is usually very deep, the training of our SA-LuT-Net may encounter the gradient vanishing issue. That is, the loss gradient is not easily propagated properly to the LuT module. To alleviate this issue, we implement a three-stage training strategy. At the first stage, the segmentation module is trained as a stand-alone model without the LuT module. At the second stage, an alternative and iterative training approach is applied on the LuT and segmentation modules. To be more specific, in one epoch the LuT module is updated while the segmentation module is fixed, and then in the next epoch, the LuT module is fixed and only the segmentation module is updated. This will be iterated for an amount of epochs to gradually train both of the LuT and the segmentation modules. Since the segmentation module is frozen during the updating of LuT module, the gradient vanishing issue in the LuT module will be mitigated. In addition, as the segmentation module is much deeper than a LuT module, it requires more rounds of updating before convergence. Therefore, after the LuT module is fully trained and becomes stable in the second stage, it will be set frozen at the third stage and only the segmentation module keeps updating for a few more rounds. After all these three stages, our SA-LuT-Net completes its training process.

D. Transferring LuTs

It is of interest to have an insight of our learnt intensity adjustment: whether it is general to the segmentation task or specific to the given segmentation model. We therefore investigate how well the learnt LuTs can be transferred from one segmentation model to another. Specifically, we first obtain the LuTs learnt based on the modified 3D Unet/DMFNet, apply them to transforming the MR images, and directly use these transformed images to train the other segmentation model

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

DMFNet/modified 3D Unet correspondingly for brain tumor segmentation. In this way, we can evaluate whether the learnt LuTs could be general, to some extent, to the segmentation task. If the transferred LuTs still increase the segmentation accuracy, the generality of our LuTs will be demonstrated. Also, as the LuT module and segmentation modules are not end-to-end trained in this transferring approach, we could verify the importance of joint training in our SA-LuT-Net. The results are discussed in section III-B.5.

E. Remarks

1) Difference From Attention: It is noteworthy that the proposed sample-adaptive LuT framework is essentially different from the attention mechanism used in some image segmentation models [36]–[38]. First, they are applied in different spaces. The attention-based methods weigh the importance of different spatial regions in the convolutional feature maps, while our method highlights the segmentation-related intensity contrast in the intensity space. For example, attention commonly assigns the similar weights to spatially neighboring pixels. In contrast, our LuT transforms the pixels of a given intensity level in the same manner, even if they are spatially distant. Second, attention is usually integrated into the deep layers of the CNNs after the convolutional feature representations have been extracted. In contrast, our sample-adaptive LuT is applied at the very early layers of the deep model to preserve the critical visual clues from the beginning. Moreover, attention and our sample-adaptive LuT do not compete with each other, but could be applied together to promote segmentation from different perspectives.

2) Difference From Image-to-Image Translation: Using the learnt parameters of a LuT, we could construct the intermediate LuT-transformed image $\hat{\mathbf{x}}$ from the input image $\mathbf{x}$. Please note that this learning process is essentially different from the common image-to-image translation in at least two significant aspects. First, in common image-to-image translation, the images in the target modality (i.e., the ground-truth $\mathbf{y}$ of $\hat{\mathbf{x}}$ in our case) are required either in paired or unpaired settings during training. However, in our approach, we do not need any auxiliary information, say, the ground-truth $\mathbf{y}$. The LuT-transformed $\hat{\mathbf{x}}$ is automatically and implicitly learnt under the guide of the original segmentation task. Second, common image-to-image translation methods learn a unified image-to-image mapping: $\mathcal{M} : \mathbf{X} \rightarrow \mathbf{Y}$ by optimally fitting all training samples, where $\mathbf{X}$ is the input image domain, and $\mathbf{Y}$ is the target image domain. However, in our approach, each input sample $\mathbf{x}$ has its own mapping $\mathcal{M}_{\theta(\mathbf{x})} : \mathbf{x} \rightarrow \mathbf{y}$, governed by the the sample-specific parameters $\theta(\mathbf{x})$ of the mapping M.

III. Experimental Results

A. Dataset and Training Settings

We evaluate the proposed SA-LuT-Nets framework on BRATS2018 and BRATS2019 [6] datasets. The former consists of a training set with 285 subjects and a validation set with 66 subjects, and the later contains 335 subjects in its training set and 125 subjects in its validation set. Each subject has four-modality $240 \times 240 \times 155$ MR images, i.e., T1-weighted (T1), Fluid Attenuated Inversion Recovery (FLAIR), T2-weighted (T2), and post-contrast T1-weighted (T1ce). For the training sets, every subject also contains its corresponding segmentation ground-truths including the whole tumor (WT), the tumor core (TC), and the enhancing tumor (ET) used as three different tumor regions. For the validation sets, since the segmentation ground truth is not given, its results need to be evaluated on BRATS2018 and BRATS2019 online server. We separately apply the preprocessing steps of the two backbone models on the input MR images and then use the proposed SA-LuT-Nets for the LuT transformation and the final segmentation. The learning rates of the LuT module and the segmentation module are fixed as 0.001 and 0.0001, respectively, in our framework. The second and third training stages separately have 150 and 250 training epochs.

B. Experimental Results and Discussion

1) Comparison With Baselines: We study the effectiveness of the proposed SA-LuT-Nets framework with both the piece-wise linear mapping function and the power mapping function, and compare them with the two baseline models, i.e., the modified 3D Unet [17] and DMFNet [19], respectively. For the piece-wise linear function, we employ a three-segment piece-wise linear function in this experiment. In this comparison, the experiments are conducted for a single modality FLAIR segmentation task. For evaluation, the BRATS2018 training set is separated as five folds to cross validate. Table I reports the Dice score results of these compared methods on three tumor regions. The higher values of dice scores indicate better segmentation results.

As can be seen, our SA-LuT-Nets with both LuT mapping functions achieve overall significant improvements over their corresponding baselines. Especially, SA-LuT-Net (piece-wise linear function) based on 3D Unet increases 1.8% on WT, 5.4% on TC, and 4.4% on ET from 3D Unet, meanwhile SA-LuT-Net (piece-wise linear function) based on DMFNet improves 2.0% on WT, 9.6% on TC, and 0.8% on ET from DMFNet. We can also find that the improvement on segmenting TC region is the most salient, by learning the proposed sample-adaptive LuTs based on both two backbone models. Further comparing the results obtained by different LuT mapping functions, we can see that the three-segment piece-wise linear mapping function performs better than the power function. Compared with the power function that has only one parameter, the three-segment linear function has four learnt parameters, which has more flexibility for intensity adjustment and therefore could better negotiate with the segmentation module to improve segmentation results. Two visual examples with these two mapping functions are separately provided in Figs. 5 and 6, showing the learnt LuTs adaptively varing with the input MR images and enhancing their tumor regions to help the segmentation task.

2) Comparison With Intensity Normalization: To study the effectiveness of our end-to-end learnt LuTs, we compare the proposed SA-LuT-Net (DMFNet based) using the three-segment piece-wise linear function with the traditional intensity normalization approach [22] as the preprocessing of MR images for the FLAIR segmentation task via the five-fold cross-validation experiments on BRATS2018 training

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I

DICE SCORES OF FLAIR SEGMENTATION RESULTS ON BRATS2018 TRAINING, REPORTED BY MEAN(STD). PAIRED T-TESTS ARE CONDUCTED BETWEEN THE PROPOSED SA-LUT-NETS AND THEIR CORRESPONDING BASELINES AT THE SIGNIFICANCE LEVEL OF 0.05, RESPECTIVELY. WHEN THE SEGMENTATION IMPROVEMENT IS STATISTICALLY SIGNIFICANT, THE RESULT OF THE PROPOSED SA-LUT-NET WILL BE INDICATED BY *

Methods	WT	TC	ET
Modified 3D Unet [17] (baseline)	0.8437(0.1520)	0.5912(0.2104)	0.3520(0.2342)
SA-LuT-Net (power function, 3D Unet based, ours)	0.8512(0.1093)*	0.6372(0.1766)*	0.3869(0.2554)*
SA-LuT-Net (piece-wise linear function, 3D Unet based, ours)	**0.8621(0.1258)***	**0.6450(0.1968)***	**0.3959(0.2647)***
DMFNet [19] (baseline)	0.8549(0.1031)	0.5499(0.2408)	0.3696(0.3055)
SA-LuT-Net (power function, DMFNet based, ours)	0.8601(0.1168)	0.5914(0.2419)*	0.3659(0.2806)
SA-LuT-Net (piece-wise linear function, DMFNet based, ours)	**0.8746(0.0864)***	**0.6459(0.2353)***	**0.3776(0.2944)***

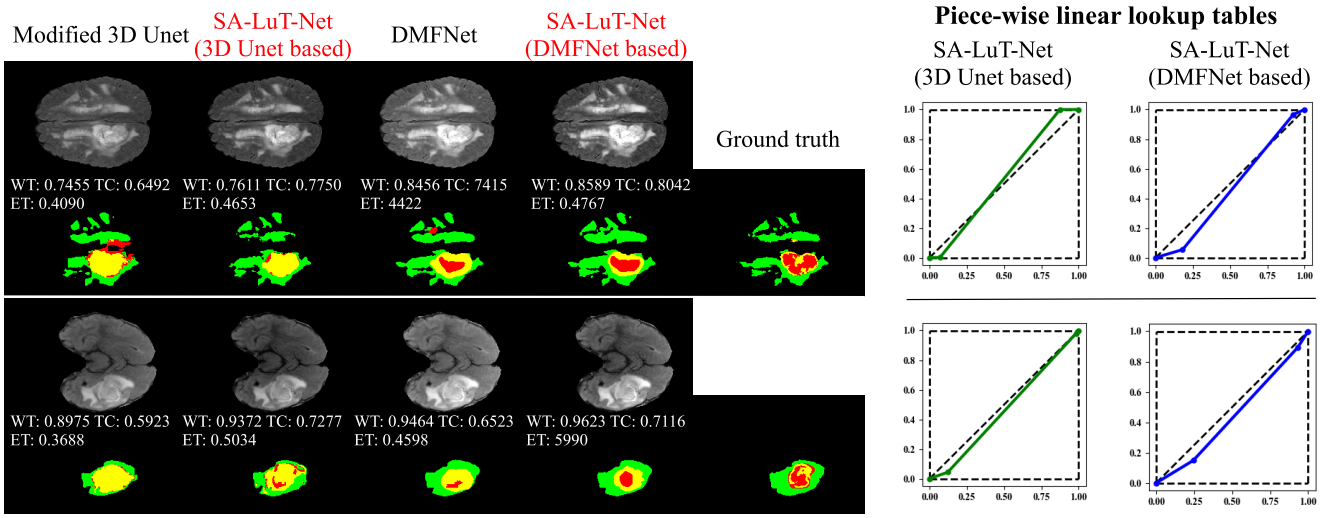

Fig. 5. Comparisons between SA-LuT-Nets using piece-wise linear mapping functions (three line segments) for LuTs and the baselines.The displayed images in first and third rows are preprocessed for the baselines and preprocessed and LuT-transformed for the SA-LuT-Nets, respectively. The second and fourth rows give their corresponding segmented labels (WT region: red+yellow+green, TC region: red+yellow, ET region: red). These learnt LuTs are flexible to adaptively adjust the intensity levels of the FLAIR MR images for the brain tumor segmentation task.

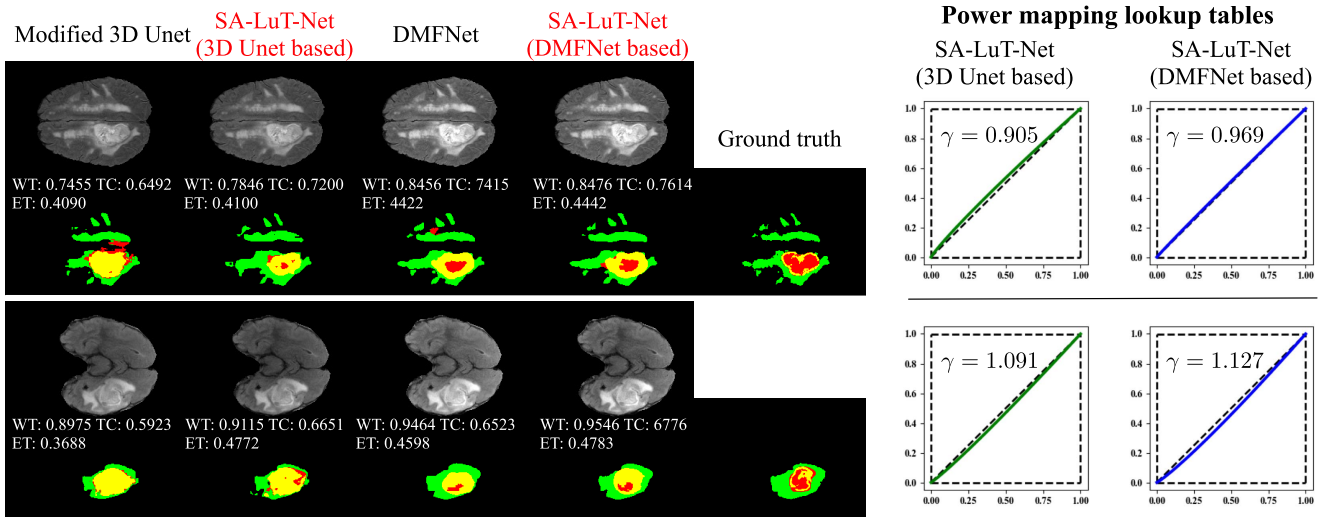

Fig. 6. Comparisons between SA-LuT-Nets using power functions for LuTs and the baselines. The displayed images in first and third rows are preprocessed for the baselines and preprocessed and LuT-transformed for the SA-LuT-Nets, respectively. The second and fourth rows give their corresponding segmented labels (WT region: red+yellow+green, TC region: red+yellow, ET region: red).

set. Their results are reported in Table II. As shown, the segmentation performance is not improved by standardizing the input MR images using the normalization approach [22].

In contrast, our end-to-end learnt LuTs based method can increase the Dice scores, which validates the effectiveness of the communication between the LuT module and the subsequent segmentation module. Because of this communication, the parameters of LuTs could be adjusted with the guide of segmentation task and help the LuT-transformed images suit the final segmentation. Two visual comparisons are shown in Fig. 7. We can see that the traditional intensity normalization [22] tends to standardize the MR images, while our intensity LuTs enhance the contrast between different tissues according to the input MR images and highlight the tumor

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISONS BETWEEN SA-LUT-NET (DMFNET BASED) AND INTENSITY NORMALIZATION APPROACH. THE DICE SCORES OF FLAIR SEGMENTATION RESULTS ARE REPORTED BY MEAN(STD)

Methods	WT	TC	ET
DMFNet [19]	0.8549(0.1031)	0.5499(0.2408)	0.3696(0.3055)
DMFNet+Norm [22]	0.8494(0.0979)	0.5204(0.3110)	0.3769(0.2909)
SA-LuT-Net (DMFNet based, ours)	**0.8746(0.0864)**	**0.6459(0.2353)**	**0.3776(0.2944)**

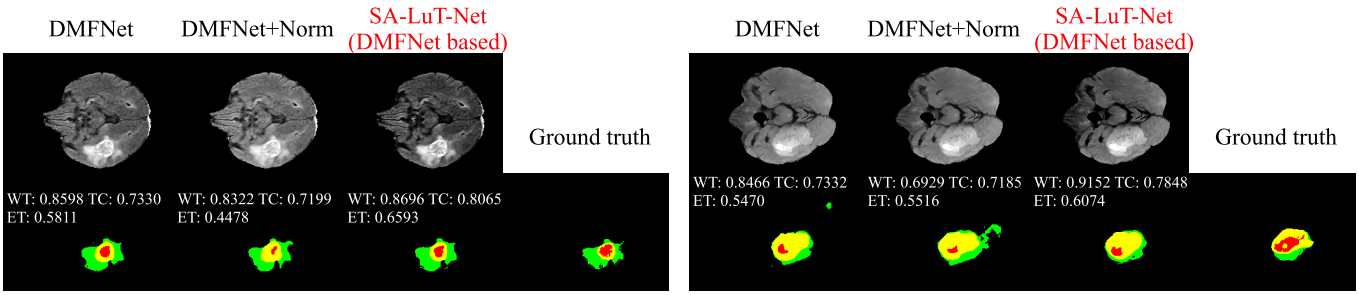

Fig. 7. Comparisons between SA-LuT-Net (DMFNet based) and intensity normalization approach [22]. The displayed images in first rows are preprocessed for the baseline and preprocessed and LuT-transformed for the SA-LuT-Net. For the intensity normalization used approach, they are standardized and preprocessed. Using the proposed end-to-end learnt LuTs, the LuT-transformed MR images have more protruded tumor regions and get better segmentation results (WT region: red+yellow+green, TC region: red+yellow, ET region: red).

information for the segmentation. Both the quantitative and the visual results demonstrate the advantages of our proposed end-to-end trained SA-LuT-Net in segmentation tasks.

3) Comparison With the State-of-the-Art: We also conducted four-modality tumor segmentation for an extensive comparison with the state-of-the-art brain tumor segmentation models on the **BRATS2018 and BRATS2019 validation sets**, respectively. We use the SA-LuT-Net (DMFNet based) with the three-segment piece-wise linear mapping in this investigation, considering that on this task DMFNet performed significantly better than the modified 3D Unet and the piece-wise linear function is more flexible than the power function for intensity adjustment. Two different segmentation evaluation metrics, i.e., Dice score and Hausdorff95 distance, are measured and reported by BRATS2018 and BRATS2019 online server, as other methods in comparison do. The lower value of the Hausdorff95 distance means better segmentation result.

Please note that since the proposed SA-LuT-Net is a single model, we focus on comparing with those single-model based approaches that have the leading performance on BRATS2018 and BRATS2019 datasets. We reported the performance of the compared methods documented in their published papers and exclude those appearing on the leaderboard but without proper documentation due to the lack of essential method details for a fair comparison.

a) Results on BRATS2018 validation set: Table III presents the results on BRATS2018. Among the compared methods, NVDLMED [16] and No New-Net [18] are the first- and second-place winners in BRATS2018 challenge. Since NVDLMED has both ensemble- and single-model versions, for a fair comparison, the performance of its single-model version is quoted here from [16]. The results of other compared single models are quoted from the published papers [18]–[20], [39], [40]. It can be seen that that our SA-LuT-Net (DMFNet based) performs best among all the compared models in the segmentation of WT and TC tumor regions in terms of the two evaluation metrics. It again achieves better segmentation results than its baseline DMFNet [19] on all the three regions. The improvement is especially significant on TC tumor region with 2.9% Dice score. On ET region, the proposed SA-LuT-Net attains slightly inferior performance to NVDLMED [16] and No New-Net [18]. Whereas, the proposed SA-LuT-Net has much fewer parameters (6.14M) to learn. This is in contrast to NVDLMED and No New-Net, which have 40.06M and 10.36M parameters, respectively, as calculated in [19]. Also, the proposed SA-LuT-Net has 27.98G floating point operations per second (FLOPs) as calculated by flops-counter.pytorch, which is significantly fewer than 202.25G (No New-Net) and 1495.53G (NVDLMED) [19]. Overall, our SA-LuT-Net achieves better performance than the state-of-the-art single models, and uses relatively fewer learning parameters. Fig. 8 gives a visual example, where the LuTs are learnt differently varying with the four MRI modalities. With the transformation by the learnt LuTs, the tumor tissues are more contrasted, and are better discriminated by the subsequent segmentation network.

b) Results on BRATS2019 validation set: For BRATS2019 validation set, since the aforementioned models did not formally present their results and the first- and third-place winners of BRATS2019 use ensemble models, we compare the proposed SA-LuT-Net based on DMFNet with four well-performed single models including the second-place winner [44] in the challenge. Their results are quoted in Table IV. Again, SA-LuT-Net (DMFNet based) outperforms its baseline DMFNet on three regions. Also, although our SA-LuT-Net performs slightly worse than the model from [44] on WT, it improves [44] by 1.3% on TC and 2.8% on ET regions. In sum, the superiority of our proposed SA-LuT-Net (DMFNet based) over the compared models has been consistently verified on BRATS2019 validation set.

4) Ablation Study: An ablation study is conducted to investigate how the number of line segments used in piece-wise linear mapping LuTs affects the segmentation performance. In this experiment, the single modality FLAIR segmentation task is

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

TABLE III

MULTI-MODALITY TUMOR SEGMENTATION RESULTS ON **BRATS2018 VALIDATION SET**. PAIRED T-TESTS ARE CONDUCTED BETWEEN THE PROPOSED SA-LUT-NET AND ITS BASELINE, I.E., DMFNET, AT THE SIGNIFICANCE LEVEL OF 0.05, RESPECTIVELY. WHEN THE SEGMENTATION IMPROVEMENT IS STATISTICALLY SIGNIFICANT, THE RESULT OF THE PROPOSED SA-LUT-NET WILL BE INDICATED BY *

Methods	Dice scores			Hausdorff95		
	WT	TC	ET	WT	TC	ET
S3D-UNet [20]	0.8935	0.8309	0.7493	-	-	-
Wang et al. [39]	0.9021	0.8583	0.7972	6.18	6.37	3.13
Kao et al. [40]	0.9047	0.8135	0.7875	4.32	7.56	3.81
DMFNet [19] (baseline)	0.9062	0.8454	0.8012	4.66	6.44	3.06
No New-Net [18]	0.9083	0.8544	0.8101	4.27	6.52	**2.41**
NVDLMED [16]	0.9068	0.8602	**0.8173**	4.52	6.85	3.82
SA-LuT-Net (DMFNet based, ours)	**0.9116***	**0.8746***	0.8073	**3.84***	**5.16***	3.67

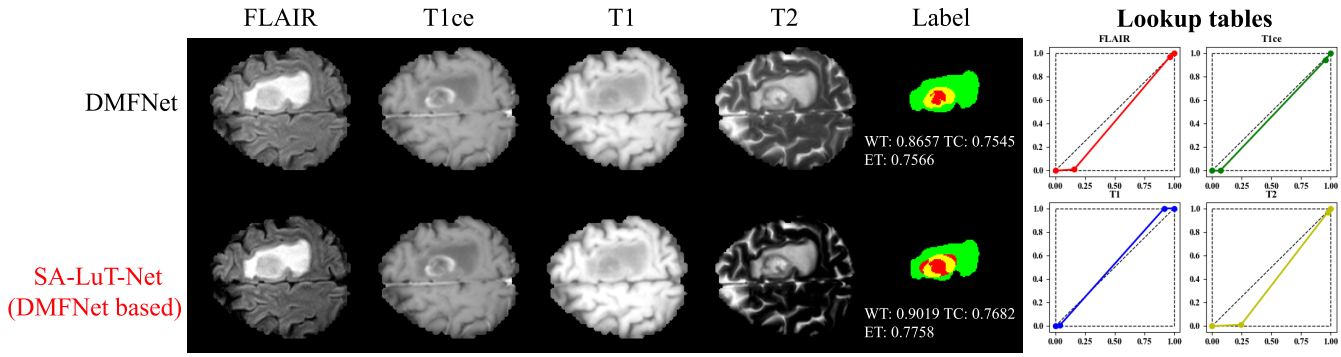

Fig. 8. A qualitative example from BRATS2018 validation set. The displayed images are preprocessed for the DMFNet and preprocessed and LuT-transformed for the proposed SA-LuT-Net. The LuT curves are learnt differently for the four-modality images. The tumor tissues are more protruded, and more easily recognized by the segmentation network after the LuT transformation in the proposed SA-LuT-Net framework (WT region: red+yellow+green, TC region: red+yellow, ET region: red).

TABLE IV

MULTI-MODALITY TUMOR SEGMENTATION RESULTS ON **BRATS2019 VALIDATION SET**. PAIRED T-TESTS ARE CONDUCTED BETWEEN THE PROPOSED SA-LUT-NET AND ITS BASELINE, I.E., DMFNET, AT THE SIGNIFICANCE LEVEL OF 0.05, RESPECTIVELY. WHEN THE SEGMENTATION IMPROVEMENT IS STATISTICALLY SIGNIFICANT, THE RESULT OF THE PROPOSED SA-LUT-NET WILL BE BE INDICATED BY *

Methods	Dice scores			Hausdorff95		
	WT	TC	ET	WT	TC	ET
Wang et al. [41]	0.8940	0.8070	0.7370	5.68	7.36	5.99
Li et al. [42]	0.8860	0.8130	0.7710	6.23	7.41	6.03
Xue et al. [43]	0.9000	0.8300	0.7500	6.13	6.77	5.07
Zhao et al. [44]	**0.9100**	0.8350	0.7540	4.57	5.58	3.84
DMFNet [19] (baseline)	0.9000	0.8018	0.7706	5.22	7.52	**3.30**
SA-LuT-Net (DMFNet based, ours)	0.9079*	**0.8482***	**0.7821***	**4.46***	**5.26***	3.69

TABLE V

THE EFFECT OF LUT LINE-SEGMENT NUMBERS USING SA-LUT-NET (DMFNET BASED) ON FLAIR SEGMENTATION

Methods	WT	TC	ET
Baseline	0.8602	0.5219	0.4251
Two line segments	0.8594	0.5323	0.4095
Three line segments	**0.8856**	**0.6494**	**0.4270**
Four line segments	0.8825	0.6213	0.4196

used with the training subjects of BRATS2018 randomly separated into 80% training data and 20% test data. Two-segment, three-segment and four-segment piece-wise linear functions are tested in the proposed SA-LuT-Net (DMFNet based). Their results are compared in Table V. As reported, using the same training setup, the model with three-line-segment LuTs achieves the best results. Meanwhile, the two-line-segment LuT gets similar results as the baseline but it is inferior to the LuTs employing three or four line segments. This is possibly caused by the less flexibility of the two-segment piece-wise linear mapping function that cannot sufficiently model a sought-after non-linear intensity transformation. Besides, a visual example of the results using different numbers of line segments is provided in Fig. 9. All the three curve lines show a similar shape for this MR sample, and using the three-line segments for LuTs gets the best segmentation results.

5) Study About Transferring LuTs Between Segmentation Models: Moreover, we are interested in investigating the generality of our learnt LuTs (three-segment piece-wise linear mapping) via the approach of transferring LuTs learnt by DMFNet to 3D Unet and vice versa. We use the single imaging modality FLAIR for this experiment. The Dice scores of the segmentation results via five-fold cross validation are reported in Table VI. When comparing the transferring performance in the last two rows of the table with that by non-transferring models, we can find that this transferring approach improves the segmentation performance on WT and TC regions from the baselines, 3D Unet (1.0% for WT and 3.6% for TC) and DMFNet (1.3% for WT and 6.1% for TC). As for the ET region, this approach gets worse or similar results compared

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
FLAIR SEGMENTATION RESULTS OF PIECE-WISE LINEAR FUNCTION USED TRANSFERRING LUTS, REPORTED BY MEAN(STD)

Methods	WT	TC	ET
Modified 3D Unet [17] (baseline)	0.8437(0.1520)	0.5912(0.2104)	0.3520(0.2342)
SA-LuT-Net (3D Unet based, ours)	0.8621(0.1258)	0.6450(0.1968)	0.3959(0.2647)
DMFNet [19] (baseline)	0.8549(0.1031)	0.5499(0.2408)	0.3696(0.3055)
SA-LuT-Net (DMFNet based, ours)	0.8746(0.0864)	0.6459(0.2353)	0.3776(0.2944)
Transferring LuTs from 3D Unet to DMFNet	0.8682(0.1339)	0.6109(0.2676)	0.3302(0.2930)
Transferring LuTs from DMFNet to 3D Unet	0.8537(0.1474)	0.6272(0.2059)	0.3572(0.2681)

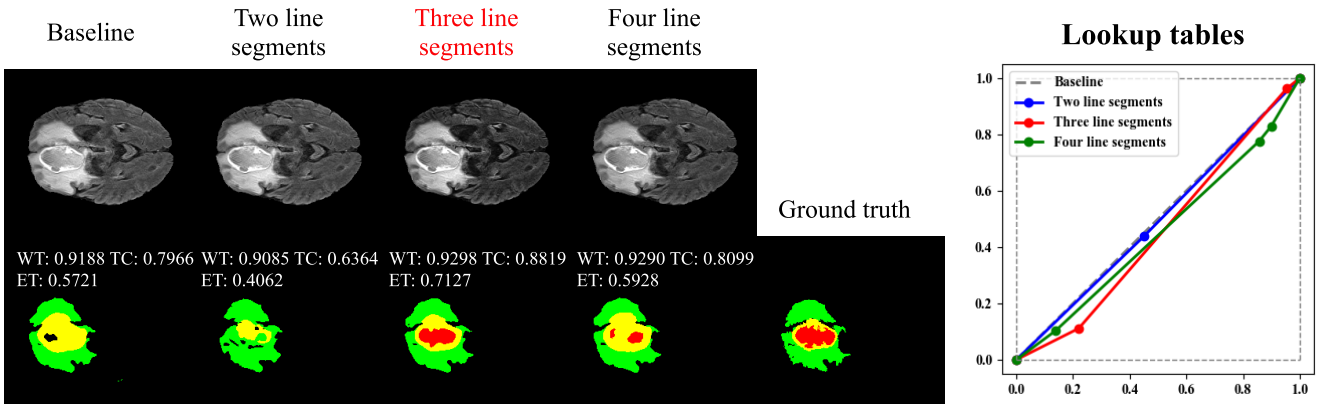

Fig. 9. A visual example of using different segment numbers. All the three curve lines show a concave shape for this MR sample, and using the three-line segments for LuTs gets the best segmentation results (WT region: red+yellow+green, TC region: red+yellow, ET region: red).

with baselines, indicating that the transferring LuTs without end-to-end training with the segmentation model may not be sufficient to depict the required contrast to segment the very small region of ET. In addition, the results of the proposed SA-LuT-Nets are better than those of the transferring approach on all the three tumor regions. This further verifies the positive effect of the end-to-end training in our SA-LuT-Net framework.

IV. DISCUSSION AND CONCLUSION

In this work, we propose to learn sample-adaptive LuTs that dynamically adjust the intensity contrast of MR images and improve the brain tumor segmentation performance. The proposed SA-LuT-Net framework, trained in an end-to-end manner, complements the novel LuT module that learns the sample-specific parameters of the nonlinear intensity mapping with a segmentation module to improve the final segmentation task. The proposed framework can be developed on various segmentation backbone networks to further improve them. **Its effectiveness was verified by the performance superiority of the proposed SA-LuT-Nets over the baselines, i.e., the modified 3D Unet and DMFNet**. The improvement purely comes from the learnt LuT transformation that enhances the task-related contrast to support the segmentation module (backbone network). The learnt LuTs acquired from the training with one segmentation network can be transferred to apply to another segmentation network, showing their relatively consistent improvements on different segmentation networks. Also, for multi-modality MR images, the proposed SA-LuT-Net generates LuTs that vary with each modality and each sample to achieve the freedom for intensity adjustment. The four-modality segmentation results by the proposed SA-LuT-Net (DMFNet based) are overall better than other state-of-the-art single models with different architectures, although the comparison with segmentation models of various structures is not the focus of this paper. It is anticipated that when more advanced brain tumor segmentation models appear, the proposed SA-LuT-Net framework could be built upon them and further improve them. In addition, the reason why the proposed three-segment piece-wise linear mapping function performs better than the power function may be similar to the reason why it outperforms the piece-wise linear function of two line segments. The power function has only one learnable parameter and the two-segment piece-wise function has two learnable parameters, giving them less freedom to adjust the intensity levels for segmentation, compared with the recommended three-segment piece-wise linear function. The four-segment piece-wise linear function gets similar results as the three-segment one, showing performance saturation on these two brain tumor segmentation datasets by using more parameters. In addition, from the experimental results, we can see that the advantage of our SA-LuT-Net is particularly salient on the tumor core part over almost all the compared models. This shows the evident benefits of the end-to-end learnt LuTs that help to highlight the essential tumor core details, while some of these details in the original MR images may not be well viewed due to relatively poor contrast.

In the future, our work can be further improved from the following perspectives. First, the effectiveness of the proposed SA-LuT-Net framework has been validated on two different backbones. We will extend it to more segmentation models in the future. Second, as mentioned, the used piece-wise linear mapping function and power function are not the only choices to transform the intensity levels. More functions depicting higher degree of nonlinearity could be explored in the proposed SA-LuT-Net framework. Third, as a general approach, the proposed learning strategy of intensity adjustment may work for other MR image analysis tasks in addition to brain

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.

tumor segmentation. Moreover, the generalization of the learnt LuTs from one analysis task to another could also be explored to further study how the requirement of image contrast could vary among these tasks.

In a conclusion, this paper proposes a framework SA-LuT-Net for brain tumor segmentation. It learns optimal sample-adaptive intensity LuTs for the given MR image segmentation task and is implemented on two prevalent backbone networks. The experimental results demonstrate that our SA-LuT-Nets improve the performance of the common segmentation models and also outperform the state-of-the-art models in the recent literature for brain tumor segmentation.

References

[1] B. H. Menze *et al.*, "The multimodal brain tumor image segmentation benchmark (BRATS)," *IEEE Trans. Med. Imag.*, vol. 34, no. 10, pp. 1993–2024, Oct. 2015.

[2] A. Isin, C. Direkoglu, and M. Sah, "Review of MRI-based brain tumor image segmentation using deep learning methods," *Procedia Comput. Sci.*, vol. 102, pp. 317–324, Dec. 2016.

[3] S. Bakas *et al.*, "Identifying the best machine learning algorithms for brain tumor segmentation, progression assessment, and overall survival prediction in the BRATS challenge," 2018, *arXiv:1811.02629*. [Online]. Available: http://arxiv.org/abs/1811.02629

[4] M. Havaei *et al.*, "Brain tumor segmentation with deep neural networks," *Med. Image Anal.*, vol. 35, pp. 18–31, Jan. 2017.

[5] M. Goetz *et al.*, "DALSA: Domain adaptation for supervised learning from sparsely annotated MR images," *IEEE Trans. Med. Imag.*, vol. 35, no. 1, pp. 184–196, Jan. 2016.

[6] S. Bakas *et al.*, "Advancing the cancer genome atlas glioma MRI collections with expert segmentation labels and radiomic features," *Sci. Data*, vol. 4, no. 1, Dec. 2017, Art. no. 170117.

[7] C. Zhou, C. Ding, X. Wang, Z. Lu, and D. Tao, "One-pass multi-task networks with cross-task guided attention for brain tumor segmentation," *IEEE Trans. Image Process.*, vol. 29, pp. 4516–4529, Feb. 2020.

[8] S. Chen, C. Ding, and M. Liu, "Dual-force convolutional neural networks for accurate brain tumor segmentation," *Pattern Recognit.*, vol. 88, pp. 90–100, Apr. 2019.

[9] M. H. Vu, T. Nyholm, and T. Löfstedt, "TuNet: End-to-end hierarchical brain tumor segmentation using cascaded networks," 2019, *arXiv:1910.05338*. [Online]. Available: http://arxiv.org/abs/1910.05338

[10] X. Zhao, Y. Wu, G. Song, Z. Li, Y. Zhang, and Y. Fan, "A deep learning model integrating FCNNs and CRFs for brain tumor segmentation," *Med. Image Anal.*, vol. 43, pp. 98–111, Jan. 2018.

[11] G. Wang, W. Li, S. Ourselin, and T. Vercauteren, "Automatic brain tumor segmentation using cascaded anisotropic convolutional neural networks," in *Proc. Int. MICCAI Brainlesion Workshop*. Quebec City, QC, Canada: Springer, 2017, pp. 178–190.

[12] G. Litjens *et al.*, "A survey on deep learning in medical image analysis," *Med. Image Anal.*, vol. 42, pp. 60–88, Dec. 2017.

[13] S. Pereira, A. Pinto, V. Alves, and C. A. Silva, "Brain tumor segmentation using convolutional neural networks in MRI images," *IEEE Trans. Med. Imag.*, vol. 35, no. 5, pp. 1240–1251, May 2016.

[14] K. Kamnitsas *et al.*, "Efficient multi-scale 3D CNN with fully connected CRF for accurate brain lesion segmentation," *Med. Image Anal.*, vol. 36, pp. 61–78, Feb. 2017.

[15] O. Ronneberger, P. Fischer, and T. Brox, "U-net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Munich, Germany: Springer, 2015, pp. 234–241.

[16] A. Myronenko, "3D MRI brain tumor segmentation using autoencoder regularization," in *Proc. Int. MICCAI Brainlesion Workshop*. Granada, Spain: Springer, 2018, pp. 311–320.

[17] F. Isensee, P. Kickingereder, W. Wick, M. Bendszus, and K. H. Maier-Hein, "Brain tumor segmentation and radiomics survival prediction: Contribution to the brats 2017 challenge," in *Proc. Int. MICCAI Brainlesion Workshop*. Quebec City, QC, Canada: Springer, 2017, pp. 287–297.

[18] F. Isensee, P. Kickingereder, W. Wick, M. Bendszus, and K. H. Maier-Hein, "No new-net," in *Proc. Int. MICCAI Brainlesion Workshop*. Granada, Spain: Springer, 2018, pp. 234–244.

[19] C. Chen, X. Liu, M. Ding, J. Zheng, and J. Li, "3D dilated multi-fiber network for real-time brain tumor segmentation in MRI," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.* Shenzhen, China: Springer, 2019, pp. 184–192.

[20] W. Chen, B. Liu, S. Peng, J. Sun, and X. Qiao, "S3D-UNet: Separable 3D U-Net for brain tumor segmentation," in *Proc. Int. MICCAI Brainlesion Workshop*. Granada, Spain: Springer, 2018, pp. 358–368.

[21] L. G. Nyál and J. K. Udupa, "On standardizing the MR image intensity scale," *Magn. Reson. Med.*, vol. 42, no. 6, pp. 1072–1081, Dec. 1999.

[22] L. G. Nyul, J. K. Udupa, and X. Zhang, "New variants of a method of MRI scale standardization," *IEEE Trans. Med. Imag.*, vol. 19, no. 2, pp. 143–150, Feb. 2000.

[23] M. Shah *et al.*, "Evaluating intensity normalization on MRIs of human brain with multiple sclerosis," *Med. Image Anal.*, vol. 15, no. 2, pp. 267–282, Apr. 2011.

[24] G. Collewet, M. Strzelecki, and F. Mariette, "Influence of MRI acquisition protocols and image intensity normalization methods on texture classification," *Magn. Reson. Imag.*, vol. 22, no. 1, pp. 81–91, Jan. 2004.

[25] D. C. Castro and B. Glocker, "Nonparametric density flows for MRI intensity normalisation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Springer, 2018, pp. 206–214.

[26] H. Lee, M. Kim, and S. Do, "Practical window setting optimization for medical image deep learning," 2018, *arXiv:1812.00572*. [Online]. Available: http://arxiv.org/abs/1812.00572

[27] A. Hoogi *et al.*, "Adaptive local window for level set segmentation of CT and MRI liver lesions," *Med. Image Anal.*, vol. 37, pp. 46–55, Apr. 2017.

[28] B. Yu *et al.*, "Learning sample-adaptive intensity lookup table for brain tumor segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Springer, 2020, pp. 216–226.

[29] M. Ju, C. Ding, Y. J. Guo, and D. Zhang, "IDGCP: Image dehazing based on gamma correction prior," *IEEE Trans. Image Process.*, vol. 29, pp. 3104–3118, 2020.

[30] H. Farid, "Blind inverse gamma correction," *IEEE Trans. Image Process.*, vol. 10, no. 10, pp. 1428–1433, Oct. 2001.

[31] S.-C. Huang, F.-C. Cheng, and Y.-S. Chiu, "Efficient contrast enhancement using adaptive gamma correction with weighting distribution," *IEEE Trans. Image Process.*, vol. 22, no. 3, pp. 1032–1041, Mar. 2013.

[32] F. Kallel and A. Ben Hamida, "A new adaptive gamma correction based algorithm using DWT-SVD for non-contrast CT image enhancement," *IEEE Trans. Nanobiosci.*, vol. 16, no. 8, pp. 666–675, Dec. 2017.

[33] Y. Liang, L. Yang, and H. Fan, "Image enhancement for liver CT images," *Proc. SPIE Int. Soc. Opt. Photon.*, vol. 7513, Nov. 2009, Art. no. 75130K.

[34] K. Somasundaram and P. Kalavathi, "Medical image contrast enhancement based on gamma correction," *Int. J. Knowl. Manage. E-Learn.*, vol. 3, no. 1, pp. 15–18, 2011.

[35] M. Sahnoun, F. Kallel, M. Dammak, C. Mhiri, K. Ben Mahfoudh, and A. Ben Hamida, "A comparative study of MRI contrast enhancement techniques based on traditional gamma correction and adaptive gamma correction: Case of multiple sclerosis pathology," in *Proc. 4th Int. Conf. Adv. Technol. Signal Image Process. (ATSIP)*, Mar. 2018, pp. 1–7.

[36] H. Xu, H. Xie, Y. Liu, C. Cheng, C. Niu, and Y. Zhang, "Deep cascaded attention network for multi-task brain tumor segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Shenzhen, China: Springer, 2019, pp. 420–428.

[37] Y. Wang, Y. Zhou, W. Shen, S. Park, E. K. Fishman, and A. L. Yuille, "Abdominal multi-organ segmentation with organ-attention networks and statistical fusion," *Med. Image Anal.*, vol. 55, pp. 88–102, Jul. 2019.

[38] Q. Jin, Z. Meng, C. Sun, L. Wei, and R. Su, "RA-UNet: A hybrid deep attention-aware network to extract liver and tumor in CT scans," 2018, *arXiv:1811.01328*. [Online]. Available: http://arxiv.org/abs/1811.01328

[39] A. Sankari and S. Vigneshwari, "Automatic tumor segmentation using convolutional neural networks," in *Proc. 3rd Int. Conf. Sci. Technol. Eng. Manage. (ICONSTEM)*, Mar. 2017, pp. 61–72.

[40] P.-Y. Kao, T. Ngo, A. Zhang, J. W. Chen, and B. Manjunath, "Brain tumor segmentation and tractographic feature extraction from structural mr images for overall survival prediction," in *Int. MICCAI Brainlesion Workshop*. Granada, Spain: Springer, 2018, pp. 128–141.

[41] F. Wang, R. Jiang, L. Zheng, C. Meng, and B. Biswal, "3D U-Net based brain tumor segmentation and survival days prediction," 2019, *arXiv:1909.12901*. [Online]. Available: http://arxiv.org/abs/1909.12901

[42] X. Li, G. Luo, and K. Wang, "Multi-step cascaded networks for brain tumor segmentation," 2019, *arXiv:1908.05887*. [Online]. Available: http://arxiv.org/abs/1908.05887

[43] Y. Xue *et al.*, "A multi-path decoder network for brain tumor segmentation," *Proc. BraTS*, 2019, pp. 255–265.

[44] Y. Zhao, Y. Zhang, M. Song, and C. Liu, "Multi-view semi-supervised 3D whole brain segmentation with a self-ensemble network," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent*. Shenzhen, China: Springer, 2019, pp. 256–265.

Authorized licensed use limited to: National University Fast. Downloaded on November 03,2023 at 15:18:10 UTC from IEEE Xplore. Restrictions apply.